

A random variable

A random variable (RV), usually written X , assigns a numerical value to each outcome of an experiment.

For two coin tosses, how could X count the heads?

Variable or realization?

Let X be the number of heads in two independent fair-coin tosses. The observed result is HT.

What is the realized value x ?

The possible values

X is the number of heads in three coin tosses.

What values can X take?

Discrete versus continuous

X counts downloaded files; Y measures a download's duration.

Which variable is discrete?

Discrete random variable

A discrete random variable has a finite or countably infinite set of possible numerical values.

Can a discrete random variable have infinitely many possible values?

Outcomes versus values

For two independent fair-coin tosses, X counts heads.

Which sample-space outcomes produce $X = 1$?

Build a distribution

For two independent fair-coin tosses, X counts heads.

What probabilities belong to $X = 0$, 1 , and 2 ?

Equal values, unequal chances

For three independent fair-coin tosses, X counts heads.

Are its four possible values equally likely?

A discrete probability distribution

A discrete probability distribution function (PDF) lists values and probabilities.

x	$P(X = x)$
0	0.1
1	0.3
2	0.4
3	0.2

What does $P(X = 2)$, also written $P(2)$, mean here?

Probability requirements

Every probability must be between 0 and 1, and all value probabilities must sum to 1.

x	$P(X = x)$
0	0.1
1	0.3
2	0.4
3	0.2

Does this synthetic model satisfy both requirements?

A missing probability

Synthetic model: $P(X = 0) = 0.2$, $P(X = 1) = 0.5$; X can only be 0, 1, or 2.

What must $P(X = 2)$ be?

An invalid distribution

Synthetic proposed probabilities for $X = 0, 1, 2$ are 0.4, 0.5, and 0.3.

Why is this not a probability distribution?

At most two

Synthetic model for files downloaded in one session.

x	$P(X = x)$
0	0.1
1	0.3
2	0.4
3	0.2

What is $P(X \leq 2)$?

Fewer than two

Synthetic model for files downloaded in one session.

x	$P(X = x)$
0	0.1
1	0.3
2	0.4
3	0.2

What is $P(X < 2)$?

At least one

Synthetic model for files downloaded in one session.

x	$P(X = x)$
0	0.1
1	0.3
2	0.4
3	0.2

What is $P(X \geq 1)$?

A complement

Synthetic model for files downloaded in one session.

x	$P(X = x)$
0	0.1
1	0.3
2	0.4
3	0.2

How could a complement simplify $P(X > 0)$?

A probability interval

Synthetic model for files downloaded in one session.

x	$P(X = x)$
0	0.1
1	0.3
2	0.4
3	0.2

What is $P(1 \leq X \leq 2)$?

From data to a model

Synthetic session counts: 0 files occurs 10 times, 1 file 30 times, 2 files 40 times, 3 files 20 times.

How could these relative frequencies supply an empirical probability distribution?

Distribution versus one observation

A model assigns $P(X = 2) = 0.4$. A single session downloads 3 files.

Does this one observation contradict the model?